**Product Information** **Ultramid®**

A3UG5

04/2025 **PA66-GF25 FR(40+30)**

## Product Information

Without halogens flame retardant glass fiber reinforced injection moulding grade. The product is light colorable and provides good mechanical and electrical properties. Due to the halide free stabilization the impact on corrosion is minimized and sensitive electronic components are better protected.

## Physical form and storage

The product is supplied in the form of granules with a bulk density of approx. 0.7 g/cm³. Standard packs are bag and bulk container (octagonal IBC=intermediate bulk container made from corrugated board with a liner bag). Other packaging materials and shipping in road or rail silo wagons are possible by agreement. The containers should only be opened immediately before processing or drying. To ensure that the delivered product absorbs as little moisture as possible, the containers should be stored in dry rooms and always carefully closed again after partial quantities have been withdrawn. In principle, the product can be stored for a long period of time. Containers stored in cold rooms should be equalized to ambient temperature before opening in order to avoid condensation on the granules. Regardless of the storage conditions, the product should be pre-dried according to our recommendations and the machine should preferably be loaded using a closed conveyor system.

## Product safety

In case processing is done under conditions as recommended (cf. processing data sheet) melts are thermally stable and do not generate hazards by molecular degradation or the evolution of gases and vapors. Like all thermoplastic polymers the product decomposes on exposure to excessive thermal load, e.g. when it is overheated or as a result of cleaning by burning off. Further information is available from the safety data sheet.

## Note

The data contained in this publication are based on our current knowledge and experience. In view of the many factors that may affect processing and application of our product, these data do not relieve processors from carrying out their own investigations and tests; neither do these data imply any guarantee of certain properties, nor the suitability of the product for a specific purpose. Any descriptions, drawings, photographs, data, proportions, weights etc. given herein may change without prior information and do not constitute the agreed contractual quality of the product. It is the responsibility of the recipient of our products to ensure that any proprietary rights and existing laws and legislation are observed.
In order to check the availability of products please contact us or our sales agency.

BASF SE

67056 Ludwigshafen, Germany

# Ultramid® A3UG5

## Product Information

We create chemistry

| Typical values for uncoloured product at 23 °C[1)] | Test method | Unit | Values[2)] |
|---|---|---|---|
| **Properties** | | | |
| Polymer abbreviation | - | - | **PA66-GF25 FR(40+30)** |
| Density | ISO 1183 | kg/m³ | **1390** |
| Viscosity number (0.5% in 96% $H_2SO_4$) | ISO 307, 1157, 1628 | cm³/g | **165** |
| Water absorption, saturation in water at 23°C | similar to ISO 62 | % | **4 - 4.6** |
| Moisture absorption, equilibrium 23°C/50% r.h. | similar to ISO 62 | % | **1.1 - 1.5** |
| **Processing** | | | |
| Melting temperature, DSC | ISO 11357-1/-3 | °C | **260** |
| MVR 275 °C/5 kg | ISO 1133 | cm³/10min | **25** |
| Melt temperature, injection moulding/extrusion | - | °C | **280 - 300** |
| Mould temperature, injection moulding | - | °C | **80 - 90** |
| Molding shrinkage, model-housing 1.5 mm | - | % | **0.4** |
| Molding shrinkage (parallel) | ISO 294-4 | % | **0.40** |
| Molding shrinkage (normal) | ISO 294-4 | % | **1.20** |
| **Thermal properties** | | | |
| Deflection temp. under load 1.8 MPa (HDT A) | ISO 75-1/-2 | °C | **245** |
| Deflection temp. under load 0.45 MPa (HDT B) | ISO 75-1/-2 | °C | **260** |
| Temperature limit for high temperatures, 20000 h, related to 50% decrease of tensile strength | IEC 60216 | °C | **130** |
| Temperature limit for high temperatures, 5000 h, related to 50% decrease of tensile strength | IEC 60216 | °C | **155** |
| Coeff. of linear therm. expansion 23°C - 55°C (parallel) | ISO 11359-1/-2 | E-6/K | **28** |
| Coeff. of linear therm. expansion 23°C - 55°C (normal) | ISO 11359-1/-2 | E-6/K | **72** |
| **Flammability (UL-yellow card see attachment)** | | | |
| GWFI (thickness) | IEC 60695-2-12 | °C (mm) | **960 (0.75)** |
| GWIT (thickness) | IEC 60695-2-13 | °C (mm) | **775 (≥ 2)** |
| Railway: Hazard level acc. to requ. sets R22 and R23 | EN 45545-2 | class | **HL3 (0,8 - 6mm)** |
| **Electrical properties** | | | **dry / cond.** |
| Relative permittivity (1 MHz) | IEC 62631-2-1 | - | **3.8 / 4.6** |
| Dissipation factor (1 MHz) | IEC 62631-2-1 | E-4 | **170 / 1000** |
| Volume resistivity | IEC 62631-3-1 | Ohm*m | **1E13 / 1E10** |
| Surface resistivity | IEC 62631-3-2 | Ohm | **- / 1E13** |
| CTI, solution A | IEC 60112 | - | **600** |
| Electric strength K20/K20, (60*60*1 mm³) | IEC 60243-1 | kV/mm | **40 / 37** |
| **Mechanical properties** | | | **dry / cond.** |
| Tensile modulus | ISO 527-1/-2 | MPa | **9500 / 6100** |
| Stress at break | ISO 527-1/-2 | MPa | **145 / 90** |
| Strain at break | ISO 527-1/-2 | % | **3 / 5** |
| Flexural modulus | ISO 178 | MPa | **9500 / 6100** |
| Flexural strength | ISO 178 | MPa | **230 / 160** |
| Charpy unnotched impact strength, 23°C | ISO 179/1eU | kJ/m² | **65 / 65** |
| Charpy unnotched impact strength, -30°C | ISO 179/1eU | kJ/m² | **63 / -** |
| Charpy notched impact strength, 23°C | ISO 179/1eA | kJ/m² | **7.5 / 9** |

Footnotes
1) If product name or properties don't state otherwise.
2) The asterisk symbol '*' signifies inapplicable properties.

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# Ultramid® A3UG5

## UL - Yellow Card

Component - Plastics E41871

BASF SE

Performance Materials Europe, PMD/EX - H201, Ludwigshafen 67056 DE

A3UG5(t)(f2)

Polyamide 66 (PA66) "Ultramid", furnished as pellets

| Color | Min. Thk (mm) | Flame Class | HWI | HAI | RTI Elec (°C) | RTI Imp (°C) | RTI Str (°C) |
|---|---|---|---|---|---|---|---|
| ALL | 0.75 | V-0 | 0 | 0 | 120 | 90 | 130 |
| BK, GY | 1.5 | V-0, 5VA | 0 | 0 | 120 | 90 | 130 |

| | | | |
|---|---|---|---|
| Comparative Tracking Index (CTI): | 0 | Inclined Plane Tracking (IPT) kV: | - |
| Dielectric Strength (kV/mm): | 21 | Volume Resistivity ($10^x$ohm-cm): | - |
| High-Voltage Arc Tracking Rate (HVTR): | - | Surface Resistivity ($10^x$ohms/square): | - |
| Dimensional Change (%): | - | High Volt, Low Current Arc Resis (D495): | - |

(f2) - Subjected to one or more of the following tests: Ultraviolet Light, Water Exposure or Immersion in accordance with UL 746C, where the acceptability for outdoor use is to be determined by UL.

(t) - May be followed by the letters LS and a color code indicating laser sensitive coloring.

ANSI/UL 94 small-scale test data does not pertain to building materials, furnishings and related contents. ANSI/UL 94 small-scale test data is intended solely for determining the flammability of plastic materials used in the components and parts of end-product devices and appliances, where the acceptability of the combination is determined by UL.

Report Date: 2005-03-31

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Last Revised: 2019-05-21

IEC and ISO Test Methods

| Test Name | Test Method | Units | Thk (mm) | Value |
|---|---|---|---|---|
| Flammability | IEC 60695-11-10, IEC 60695-11-20 | Class (color) | 0.75 | V-0 (ALL) |
| | | | 1.5 | V-0, 5VA (BK, GY) |
| Glow-Wire Flammability (GWFI) | IEC 60695-2-12 | °C | 0.75 | 960 |
| | | | 1.5 | 960 |
| Glow-Wire Ignition (GWIT) | IEC 60695-2-13 | °C | - | - |
| IEC Comparative Tracking Index | IEC 60112 | Volts (Max) | - | - |
| IEC AC Dielectric Strength (AC DS) | IEC 60243-1 | kV/mm | - | - |
| IEC DC Dielectric Strength (DC DS) | IEC 60243-2 | kV/mm | - | - |
| IEC Volume Resistivity (VR) | IEC 62631-3-1 | 10x ohm-m | - | - |
| IEC Surface Resistivity (SR) | IEC 62631-3-2 | 10x ohms | - | - |
| IEC Inclined Plane Tracking (IPT) | IEC 60587 | kV | - | - |
| IEC Ball Pressure | IEC 60695-10-2 | °C | - | - |
| ISO Heat Deflection (1.80 MPa) | ISO 75-2 | °C | - | - |
| ISO Tensile Strength | ISO 527-2 | MPa | - | - |

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| | | | | |
|---|---|---|---|---|
| ISO Flexural Strength | ISO 178 | MPa | - | - |
| ISO Tensile Impact | ISO 8256 | kJ/m2 | - | - |
| ISO Izod Impact | ISO 180 | kJ/m2 | - | - |
| ISO Charpy Impact | ISO 179-1 | kJ/m2 | - | - |

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## UL - Yellow Card

Component - Plastics E41871

BASF SE

Performance Materials Europe, PMD/EX - H201, Ludwigshafen 67056 DE

A3UG5 (t)(f1)

Polyamide 66 (PA66) "Ultramid", furnished as pellets

| Color | Min. Thk (mm) | Flame Class | HWI | HAI | RTI Elec (°C) | RTI Imp (°C) | RTI Str (°C) |
|---|---|---|---|---|---|---|---|
| BK, GY | 0.75 | V-0 | 0 | 0 | 120 | 90 | 130 |
| | 1.5 | V-0, 5VA | 0 | 0 | 120 | 90 | 130 |

Comparative Tracking Index (CTI): 0

Inclined Plane Tracking (IPT) kV: -

Dielectric Strength (kV/mm): 21

Volume Resistivity ($10^x$ohm-cm): -

High-Voltage Arc Tracking Rate (HVTR): -

Surface Resistivity ($10^x$ohms/square): -

Dimensional Change (%): -

High Volt, Low Current Arc Resis (D495): -

(f1) - Suitable for outdoor use with respect to exposure to Ultraviolet Light, Water Exposure and Immersion in accordance with UL 746C.

(t) - May be followed by the letters LS and a color code indicating laser sensitive coloring.

ANSI/UL 94 small-scale test data does not pertain to building materials, furnishings and related contents. ANSI/UL 94 small-scale test data is intended solely for determining the flammability of plastic materials used in the components and parts of end-product devices and appliances, where the acceptability of the combination is determined by UL.

Report Date: 2005-03-31

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IEC and ISO Test Methods

| Test Name | Test Method | Units | Thk (mm) | Value |
|---|---|---|---|---|
| Flammability | IEC 60695-11-10, IEC 60695-11-20 | Class (color) | 0.75 | V-0 (BK, GY) |
| | | | 1.5 | V-0, 5VA (BK, GY) |
| Glow-Wire Flammability (GWFI) | IEC 60695-2-12 | °C | 0.75 | 960 |
| | | | 1.5 | 960 |
| Glow-Wire Ignition (GWIT) | IEC 60695-2-13 | °C | - | - |
| IEC Comparative Tracking Index | IEC 60112 | Volts (Max) | - | - |
| IEC AC Dielectric Strength (AC DS) | IEC 60243-1 | kV/mm | - | - |
| IEC DC Dielectric Strength (DC DS) | IEC 60243-2 | kV/mm | - | - |
| IEC Volume Resistivity (VR) | IEC 62631-3-1 | 10x ohm-m | - | - |
| IEC Surface Resistivity (SR) | IEC 62631-3-2 | 10x ohms | - | - |
| IEC Inclined Plane Tracking (IPT) | IEC 60587 | kV | - | - |
| IEC Ball Pressure | IEC 60695-10-2 | °C | - | - |
| ISO Heat Deflection (1.80 MPa) | ISO 75-2 | °C | - | - |
| ISO Tensile Strength | ISO 527-2 | MPa | - | - |

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## UL - Yellow Card

| | | | | |
|---|---|---|---|---|
| ISO Flexural Strength | ISO 178 | MPa | - | - |
| ISO Tensile Impact | ISO 8256 | kJ/m2 | - | - |
| ISO Izod Impact | ISO 180 | kJ/m2 | - | - |
| ISO Charpy Impact | ISO 179-1 | kJ/m2 | - | - |

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